# Generation, retrieval-augmented systems, and where this leaves you

COCO and the Part V corpora

LECTURE 24 GÉRON CH. 15–16

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

# Where we are

Lecture 23 built a catalogue you can search with a sentence. Two weak spots remain, and both are about entries rather than about the model:

- Some entries have **no usable description** — nothing for the text tower to encode
- Some queries are **ambiguous**, and a ranking is a poor answer to a question the user has not finished asking

**BOTH ARE REPAIRED BY GENERATION**

A multimodal model can *write* the missing description. And a generator with the retriever attached can answer the ambiguous query in words, grounded in what the catalogue actually contains.

# Today

1. Captioning: filling the holes in the catalogue (20 min)
2. Retrieval-augmented generation — Lecture 20’s retriever joined to Lecture 18’s generator (25 min)
3. Failures that belong to the join rather than to either half (15 min)
4. **The course, as one argument** (25 min)

THE FIRST REPAIR

# Writing the missing descriptions

20 minutes

# The hole in the index

60 of 200 catalogue entries have no description. A text index cannot rank a document that does not exist, so those entries score exactly 0% — structurally, not statistically.

> **THE IDEA**
>
> We have a model that maps images to text: Lecture 23 measured it running in the image-to-text direction. Use a model built for *generating* the text rather than retrieving it, and write the missing descriptions.

# The captioner

| Component | What it is |
|---|---|
| Image encoder | a vision transformer — Chapter 16 |
| Text decoder | a transformer decoder with cross-attention to the image — Chapter 15 |
| Training | image–caption pairs, next-token prediction |
| Size | 224 million parameters |

Architecturally this is the encoder–decoder of Chapter 15 with an image encoder in place of the text one. Nothing new; a different pair of modalities in the same box.

# The repair, in nine lines

```python
from transformers import BlipForConditionalGeneration, BlipProcessor

proc = BlipProcessor.from_pretrained("Salesforce/blip-image-captioning-base")
cap  = BlipForConditionalGeneration.from_pretrained(
    "Salesforce/blip-image-captioning-base").to(device).eval()

with torch.no_grad():
    batch = proc(images=missing_images, return_tensors="pt").to(device)
    ids   = cap.generate(**batch, max_new_tokens=30, num_beams=3)

generated = proc.batch_decode(ids, skip_special_tokens=True)
assert len(generated) == len(missing_images)
```

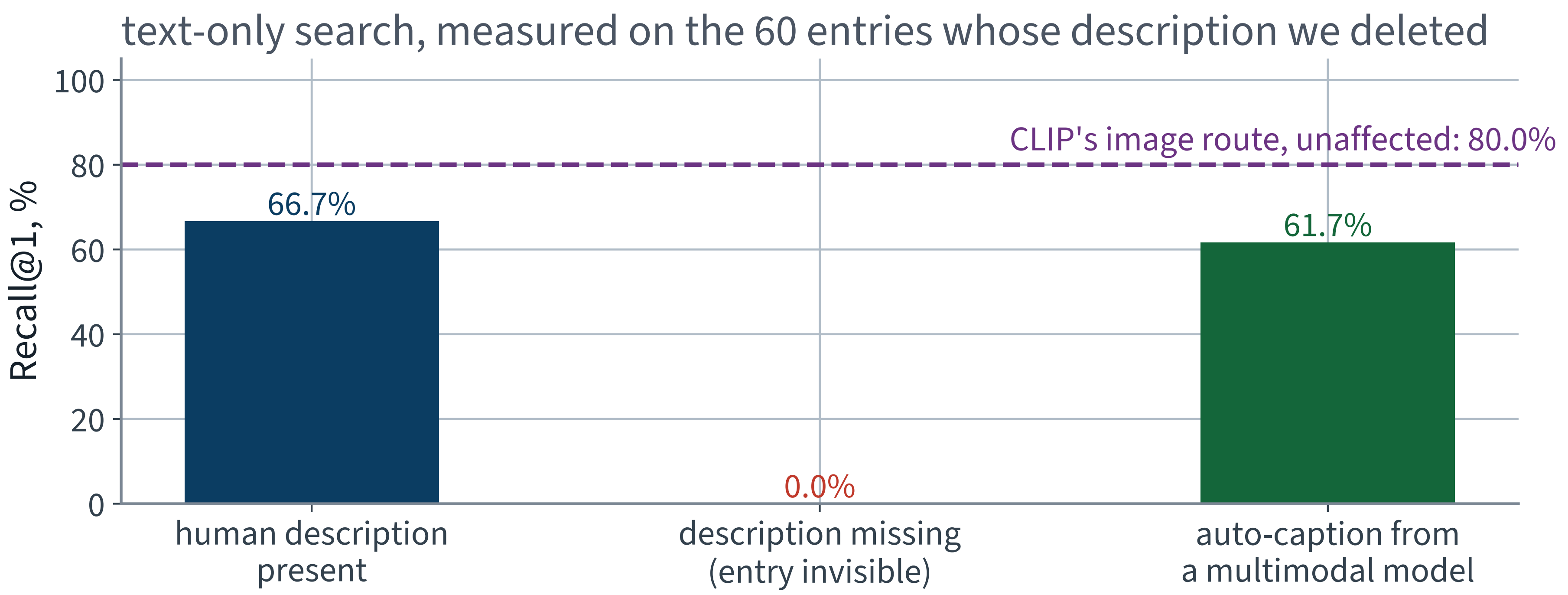
Expected wall clock: 21 s for 60 images with beam search. Not instant; run it offline, once, per new entry.

# What it writes

| Human description (deleted) | Generated description |
| --- | --- |
| This wire metal rack holds several pairs of shoes and sandals | a dog laying on top of a pair of shoes |
| A green vase filed with red roses sitting on top of table. | a vase of flowers sitting on a window sie |
| A giraffe bending over while standing on green grass. | the gi is brown and white |

The first row is the interesting one: **both sentences are true**. There is a dog asleep on the shoe rack, and the two writers chose different subjects. The third row is not a sentence at all.

# Better than nothing, measured




From 0% to 61.7% on the 60 affected entries — against 66.7% if a person had written them.

# The whole catalogue, three ways

| Text-only route, all 200 queries | R@1 | R@10 |
|---|---|---|
| Every entry described by a person | 61.5% | 96.0% |
| **✗ 60 entries with nothing** | **✗ 46.0%** | **✗ 68.5%** |
| **✓ Those 60 auto-captioned** | **✓ 62.0%** | **✓ 94.0%** |

At rank 1 the recovery is complete to within one query out of 200 — which is noise, not a win over the human writers. At rank 10 it is not complete, and that gap is the real finding.

# Meanwhile, the route that never read a description

| On the same 60 entries | R@1 |
|---|---|
| Text index, description deleted | ✗ 0% |
| Text index, auto-captioned | 61.7% |
| **✓ Joint-space image route** | **✓ 80.0%** |

THE DESIGN LESSON

Two routes that fail for different reasons are worth more than one better route. The image route is immune to missing text; the text route is immune to a photograph that does not show the distinguishing feature.

# Three things not to claim

- A generated caption is **not evidence about the product**. It is a restatement of the photograph, and it will confidently name things that are not there
- The recovery is measured on 60 entries of 200. That is a small sample and the interval around it is wide
- We are scoring generated captions against human captions of the same image — a friendly test. A customer's wording is not a COCO caption

**THE FAILURE MODE TO WATCH FOR**

An auto-caption that is wrong makes an entry *findable under the wrong query*, which is worse than unfindable. Nothing we measured today detects that.

# Captioning: what it is, and is not

A multimodal model conditioned on an image, generating text. Architecturally it is Lecture 18's decoder with cross-attention into Lecture 23's image encoder — nothing new, arranged differently.

**WHAT A GENERATED CAPTION IS**

A **plausible** description, not a true one. It describes what the model expects to see, and it will confidently name objects that are not there. For catalogue entries that had *nothing*, plausible is an improvement on absent; as a replacement for a real description it is not.

So the honest use is narrow and worth stating: fill holes, mark the filled entries as machine-written, and measure retrieval with and without them.

# Prompting is a hyperparameter

A generator's output depends on how you ask, and the difference is often larger than the difference between models.

> **WHICH MAKES IT A CHOICE, WITH ALL THAT IMPLIES**
>
> Trying five prompts and keeping the one with the best test score is choosing a hyperparameter on the test set — Lecture 2's error, in a costume that does not look like modelling at all. It has a validation set like anything else.

**✗ Report the prompt with the result. A number without the prompt that produced it is not reproducible, and prompts are rarely written down because they do not feel like code.**

# What generation does not fix

- **It does not add knowledge.** A caption is a guess from pixels; RAG is a lookup. Neither creates information the system does not have
- **It does not calibrate.** Fluent and confident is the default register regardless of whether the answer is grounded
- **It does not remove the need for evaluation.** It makes it harder: the output is text, and text has no obvious metric

That last point is the honest closing note of the technical half of this course. Every lecture until now had a number. Here you have to *construct* the number first, and constructing it well is most of the work.

# The verdict

For the first time in twelve applications, the number you committed to was probably *too low* rather than too high — because the failure this time was not in the evaluation. It was in the geometry, and the repair was somebody else's pretraining run.

> **STILL WORTH CHECKING**
>
> 72.5% over 200 candidates is not 72.5% over fifty thousand. Recall rises as the haystack shrinks, and a catalogue is a haystack that grows.

# Two things we measured and did not fix

| # | Measured last lecture | Number |
|---|---|---|
| 1 | Text search R@1 on entries with no description | ✗ 0% |
| 2 | Ambiguous queries with a single right answer | ✗ none exist |

The first is a hole in the index. The second is a mismatch between what the customer asked and what a ranked list can say.

Both are repaired in the next thirty minutes, and both are re-measured before we leave.

THE SECOND REPAIR

# Retrieval-augmented generation

25 minutes · examinable

# The join, drawn

1. The user asks a question in words
2. The **retriever** of Lecture 20 embeds it and returns the $k$ most similar catalogue entries
3. Those entries are placed in the **generator**'s context, with the question
4. The generator answers *from what it was given*, and cites which entry it used

> WHY THIS IS NOT JUST A BIGGER MODEL
>
> The knowledge lives in the index, not in the weights. Add a product and it is answerable immediately; remove one and it stops being cited. No retraining, and the system can say *where* an answer came from — which is usually the requirement that decides whether it can be deployed at all.

# What each half contributes

| | Good at | Bad at |
|---|---|---|
| **Retriever** | finding the right entries, fast, over millions | saying anything; it only ranks |
| **Generator** | fluent answers, synthesis across entries | knowing anything reliably; it will produce a plausible answer regardless |

Each covers the other's weakness, which is why the combination is worth more than either. It is also why the failures are interesting: they belong to **neither half**.

# Failures of the join

THREE, AND NONE IS VISIBLE FROM EITHER COMPONENT

**Retrieved but ignored.** The right entry is in the context and the generator answers from its own weights instead.

**Not retrieved, answered anyway.** The retriever misses, and the generator does not say so — it produces a fluent answer with nothing behind it.

**Retrieved and misread.** The entry is present and correct, and the answer misstates it.

**✗ Evaluating the retriever alone (recall@k) and the generator alone (fluency) can both look excellent while the system does all three. The join needs its own measurement.**

# Chunking, and why it decides everything

Before anything is retrieved, the corpus must be cut into units. That choice is upstream of the retriever and of the generator, and it is usually made without measurement.

- **Too small** — a chunk lacks the context that makes it answerable, and the generator gets a fragment
- **Too large** — the embedding averages several topics and matches none of them well
- **Split badly** — a sentence cut in half retrieves for neither of its halves

**✗ It is a hyperparameter. Treat it as one: vary it, hold everything else fixed, and measure recall@k on the same judgements — Lecture 19’s protocol, unchanged.**

# How many to retrieve

$k$ trades two failures against each other.

| Small $k$ | Large $k$ |
| --- | --- |
| the right entry may not be there at all | it is there, buried among distractors |
| the generator cannot recover | the generator may attend to the wrong one |

> **WHY RE-RANKING EARNS ITS COST HERE**
>
> Retrieve a generous $k$ with the cheap bi-encoder, then re-rank with the cross-encoder of Lecture 20 and pass only the top few on. High recall from the first stage, high precision from the second — which is the two-stage argument of Part V, arriving where it matters most.

# How to measure the join

- **Groundedness** — is every claim in the answer supported by a retrieved entry? Check mechanically where you can
- **Attribution** — does the cited entry actually say it? A citation that does not support its claim is worse than none
- **Abstention** — when nothing relevant is retrieved, does it decline? Measure this deliberately, with queries whose answer is genuinely absent
- **End to end** — and only then, is the answer right?

> **THE CLOSED-BOOK CONTROL**
>
> Run the same questions with the retrieval turned off. Whatever the system still answers correctly, it knew already — and the difference is what retrieval actually bought. Same shape as every ablation in this course.

# What the customer actually types

- “something to sit on”
- “gear for bad weather”
- “an animal that would suit a small flat”

A ranked list of pictures is a poor answer to any of these. The customer wants a shortlist *with reasons* — and a ranking has no place to put a reason.

Twelve such queries, written by hand and listed in the notebook. There is no dataset of ambiguous queries; inventing them and saying so is the honest option.

# Two options that do not work

OPTION A

Show the top ten pictures and let the customer decide. That is what we already have, and it is why they asked in words.

OPTION B

Ask a language model. It has never seen our catalogue, so anything it recommends is invented — and it will invent it fluently.

Retrieval-augmented generation is the composition of the two: **retrieve first, then generate strictly from what was retrieved.**

# First, decide how you will check it

“The answer is good” is not measurable in a lecture. So give every entry a stock number and require the assistant to cite them:

> **THE METRIC**
>
> A cited SKU either exists in the catalogue or it does not. Count the citations, count the ones that exist, report the fraction. No judgement call anywhere in the loop.

This measures *grounding*, not helpfulness. Say which one you measured; they are not the same thing and only one of them is cheap.

# Retrieve, then generate

```
q = unit(clip_text(["something to sit on"]))      # (1, 512)
top5 = np.argsort(-(q @ I.T)[0])[:5]              # I: catalogue images

shortlist = "\n".join(f"- {cat[j]['sku']}: {cat[j]['captions'][0]}"
                      for j in top5)

prompt = (f'Here are the catalogue entries our search returned for '
          f'"something to sit on":\n{shortlist}\n\n'
          f'Recommend the best ones, citing only SKUs from the list above. '
          f'If none fit, say so.')

answer = generate(prompt)                          # a 0.5B instruction model
cited  = re.findall(r"CAT-\d{6}", answer)
print(f"{sum(s in valid_skus for s in cited)}/{len(cited)} cited SKUs exist")
```

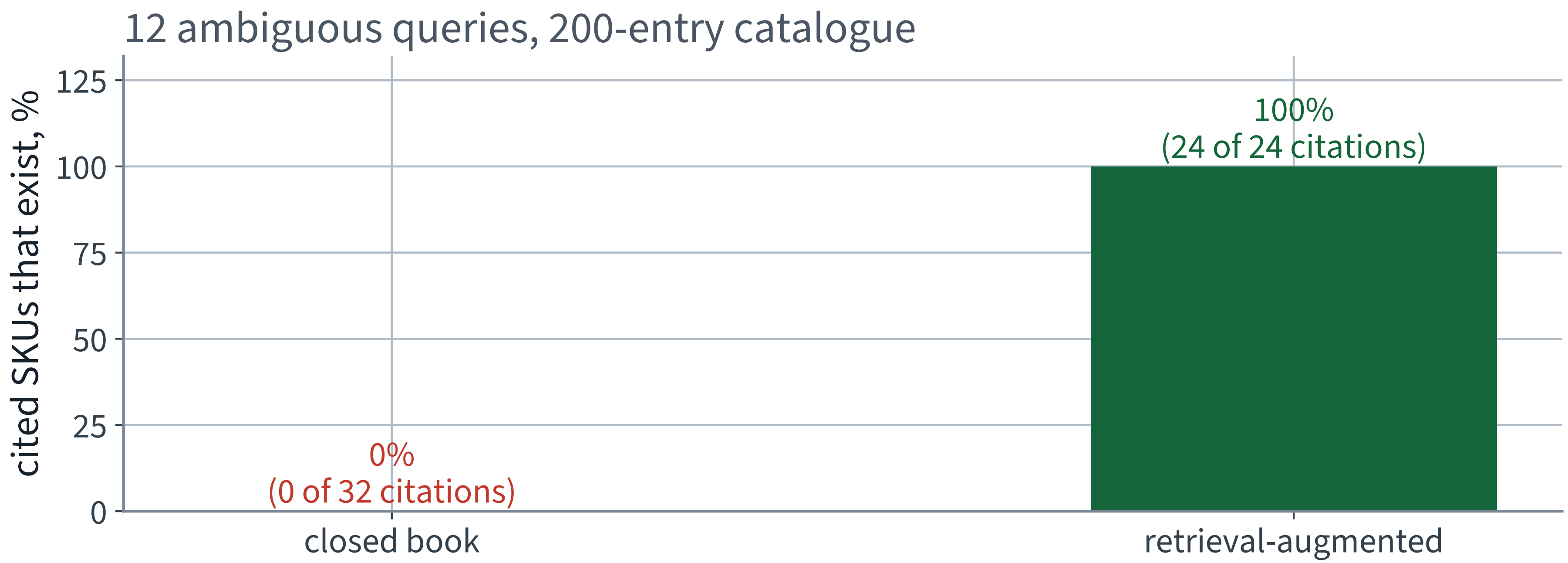
# Closed book — what it does without retrieval

**ASKED THE SAME QUESTION WITH NO CATALOGUE**

The model produces a fluent recommendation and cites stock numbers in exactly the right format. Across 12 queries it emitted 32 citations, of which **0** correspond to an entry that exists.

The failure is not that it refuses. It is that it does not refuse, and nothing in the output distinguishes an invented SKU from a real one.

# With the catalogue in the prompt




0% of closed-book citations exist against 100% when the shortlist is in the prompt — over 12 queries and a 200-entry catalogue.

# The numbers

| Over 12 ambiguous queries | Closed book | Retrieval-augmented |
| --- | ---: | ---: |
| Queries that cited at least one SKU | 10 | 12 |
| Citations emitted | 32 | 24 |
| Citations that exist | **✗ 0** | **✓ 24** |
| Grounding rate | **✗ 0%** | **✓ 100%** |

The generator is a 494-million parameter instruction model, chosen so the whole thing runs on a laptop. A larger one would follow the citation instruction more reliably; it would not change the shape of the result.

# What this does ✗ not fix

- **Grounding is not correctness.** A cited entry can exist and still be a bad recommendation. We measured the cheap half
- **The retriever is now the ceiling.** If the right entry is not in the top five, no amount of generation will recover it — which is why the R@5 we measured last lecture is the number that matters here
- **Fluency is unchanged.** A wrong answer built from real SKUs reads better than a wrong answer built from invented ones

> **WHERE THIS SITS IN THE FIELD**
>
> Evaluating generated answers — faithfulness, attribution, judging with another model — is a live research area and entirely outside Chapters 1–16. Not examinable, and worth knowing exists.

# Everything, re-measured

| Route | Before | After |
|---|---:|---:|
| Text query → image, joint space | 72.5% | 72.5% (untouched) |
| Text index, on the entries with no description | ✗ 0% | ✓ 61.7% |
| Ambiguous queries, grounding rate | ✗ 0% | ✓ 100% |

Two repairs, two measured deltas, one number deliberately unchanged as a control. That last row is what makes the other two believable.

# The notebook for this lecture

Open Lecture 24 in Colab

- **Runs on free CPU.** Small pretrained models, inference only
- Captions generated for the entries that have none, and the retrieval re-measured with them
- The closed-book control: the same questions with retrieval off, so what retrieval bought is a number

> **WHAT TO CHANGE**
>
> Ask a question whose answer is *not* in the catalogue. A system that answers it anyway has the second failure of the join, and you have just built the test that finds it.

THE LAST TWENTY-FIVE MINUTES

# The course, as one argument

25 minutes

# The argument

Every method in this course arrived because something measurable could not be done without it.

> **AND EVERY LECTURE HAD THE SAME SHAPE**
>
> A quantity that has to be derived before the method makes sense; the method; a worked example with real numbers; and the conditions under which it fails. Eighteen derivations, and they are the part that will still be true when every library in these notebooks has been replaced.

# What the first lecture promised

> **SLIDE 2, LECTURE 1**
>
> “How to build a machine learning system, and how to know whether it works. Fitting a model is a few lines. Establishing that the number it reports means what it appears to mean is the whole of the rest of the course.”

That was a claim, not a plan. Here is the evidence.

# The claim, stated precisely

Every method you met in this course was introduced at the moment a number *you produced* was wrong without it.

Not “was useful”. Not “is standard”. **Was wrong, and measurably so, and you were the one holding the wrong number.**

The rest of this block is that claim, once for each application, with the measurement attached every time.

# Thirteen datasets, fourteen first failures

## Each application, and the thing that broke first

the red line is the measurement that failed; the method arrived afterwards

1–2 · housing prices
scaled before splitting, scored on the training set
→ pipelines, cross-validation

3 · rare-event detector
90.96% accurate, and it never fires at all
→ precision, recall, thresholds

4–5 · survival probabilities
degree 5 fit the training set and nothing else
→ bias–variance, ridge, lasso

6–7 · land cover
one tree, refit, gives a different explanation
→ impurity, bagging, forests

8 · face grouping
4,096 dimensions, where light moves a face further
→ SVD, PCA, projection

9–10 · garment sorting
a loop you cannot open, and two silent bugs
→ PyTorch, autograd, eval()

11 · deep recognition
twenty layers, and the signal died on the way
→ initialisation, normalisation

12–13 · visual inspection
from scratch, on far too few labelled images
→ convolution, transfer learning

14 · object detection
a visibly wrong box, and no number for how wrong
→ IoU, GIoU, average precision

15–16 · demand forecasting
a random split across time: you forecast the past
→ stationarity, backtesting

17–18 · customer feedback
a word vocabulary the next review escapes
→ subwords, attention

19–20 · claim checking
98% of claims match no document at all
→ BM25, NDCG, dense retrieval

21–22 · recommendation
RMSE on the films they had already chosen
→ factorisation, honest protocols

23–24 · the catalogue
two encoders, two spaces, and nothing between them
→ the contrastive objective, RAG

Read the middle line of each box. That is the syllabus — the chapter order is a consequence of it.

# Part I — the ideas everything else reuses

| Lecture | The derivation | Where it came back |
| --- | --- | --- |
| 2 · end-to-end | least squares, the normal equation | ridge in 5; collinearity everywhere |
| 3 · classification | imbalance; precision is not monotone | the maximum inside AP, Lecture 14 |
| 4 · training models | gradient descent | every network from Lecture 9 on |
| 5 · regularisation | bias–variance | every learning curve since |
| 6, 7 · trees, ensembles | impurity; variance of a correlated average | multi-head attention, Lecture 18 |
| 8 · PCA | SVD; Johnson–Lindenstrauss | matrix factorisation in 21; the sphere in 23 |

# Parts II–IV — the same three ideas, deeper

| Lecture | The derivation | Where it came back |
|---|---|---|
| 9 · networks | what a layer computes | everything after it |
| 10 · PyTorch | reverse-mode autodiff | why training is affordable at all |
| 11 · deep training | variance propagation | BPTT in 16; residuals in 18 |
| 12 · convolution | weight sharing, equivariance, memory | what a ViT gives up, Lecture 23 |
| 14 · detection | IoU; mAP | ranking metrics in 19 |
| 15 · time series | stationarity; autocorrelation | causal masking in 18 |
| 17 · text | softmax, cross-entropy, logits | attention's softmax in 18; the contrastive loss in 23 |
| 18 · attention | scaled dot-product attention | Part V's encoders, and 23 |

# Part V — where the embeddings earned their living

| Lecture | The derivation | The point |
|---|---|---|
| 19 · lexical IR | MRR, AP, NDCG | a ranking is not a classification, and needs its own metrics |
| 20 · dense IR | — | two encoders and an inner product; the negatives are most of the work |
| 21 · matrix factorisation | low rank, and its relation to the SVD | Lecture 8, with missing entries |
| 22 · neural recsys | — | the two-tower model IS the bi-encoder; and offline evaluation is where recsys is usually wrong |
| 23 · multimodal | contrastive objective, temperature | the same two towers, different modalities |

Lectures 20, 22 and 23 are three uses of one object. That is the strongest single thread in the course, and it is why Part V sits after transformers rather than before.

# The eighteen derivations, in one place



**✗ Forty per cent of the written paper, and the part of this course with the longest shelf life.**

# The failures, collected

| Failure | First met | Measured cost |
|---|---|---|
| Fitting before splitting | Lecture 2 | nothing measurable — and unknowable in advance |
| Evaluating on training data | Lecture 2 | an RMSE of zero |
| Accuracy under imbalance | Lecture 3 | a useless model above 90% |
| Choosing on the test set | Lectures 2, 5 | an optimistic report, every time |
| Missing `zero_grad()` | Lecture 10 | 76.7 points |
| Default initialisation, deep | Lecture 11 | a network that does not train at all |
| Random split on a series | Lecture 15 | optimistic, and undeployable |
| Augmenting the validation set | Lecture 13 | 1.57 points of noise on the criterion |

**✗ Not one of them raises an exception. That is the whole reason the course is organised around measurement.**

# Detection, in numbers

| Detection on 128 COCO images | Value |
| --- | --- |
| Objects miscounted per image, at the 0.5 reporting threshold | **✗ 3.0** |
| …at the threshold that minimises error on these same images | 1.92 — a diagnostic, not a result |
| mAP at IoU 0.5 | 0.659 |
| mAP averaged over IoU 0.5 to 0.95 | 0.439 |
| **✗ The same detections, averaged per image** | **✗ 0.817** |

The last row is the same detections as the third, both at IoU 0.5. Averaging over images instead of over classes moves that headline by 0.157 — a mean of a mean is not a mean. Against the stricter 0.5:0.95 row directly above it the apparent gap is 0.378, and that one is not a comparison at all: the two rows are averaged over different thresholds.

# Recommendation, in numbers

| Same model, same rows, different protocol | MAE |
|---|---|
| Random cross-validation split | **44,925 ✗** |
| Honest, time-ordered split | **56,446 ✓** |
| **✗ Optimism from shuffling time** | **✗ 25.6%** |

Nothing was fitted on the test set. No preprocessing crossed the split. The leak was the *ordering*, and with positive autocorrelation a point's own near-future was in its training set.

# Six failures, and where they live outside this course

| Failure | Met in | Where it recurs |
|---|---|---|
| Leakage from fitting before splitting | L2 | any target encoding, oversampling, feature selection or imputation done on the full frame |
| Accuracy under imbalance | L3 | fraud, medical screening, click prediction, defect detection — anywhere the interesting class is rare |
| Variance mistaken for signal | L6, L8 | a single-seed comparison; a leaderboard whose top ten are within noise of each other |
| Signal vanishing with depth | L11 | any deep stack without residual connections or normalisation, including very long sequences |
| The wrong metric | L19, L14, L23 | a ranked list scored with accuracy; a detector scored without IoU; a recall quoted without its candidate set |
| Temporal leakage | L15, L22 | any user-level split where the same user appears in both halves; any shuffled panel data |

# The catalogue of silent failures, closed

| It runs, and produces a plausible number | Diagnosed in |
|---|---|
| Scaler fitted before the split | L2 |
| Evaluating on training data | L2 |
| Accuracy under class imbalance | L3 |
| Hyperparameters chosen on the test set | L2, L5, and again on a prompt template in L23 |
| Missing `model.eval()` / `optimizer.zero_grad()` | L10 |
| Metric averaged per batch, not over the set | L10 |
| Random split on an autocorrelated series | L15 |
| Embeddings compared without normalising | L23 |

**✗ Every row of that table runs to completion. Not one of them raises.**

# What did not change, in twenty-four lectures

1. **Split before anything is fitted.** Lecture 2, and every lecture since
2. **Compute the trivial baseline first.** A metric with nothing beside it is decoration
3. **Two numbers measured on different rows are not comparable.** However similar the column headings look
4. **A difference smaller than the spread is not a difference.** Report the spread
5. **Change one thing.** Otherwise the difference is not attributable to anything

**✗ Five rules, no mathematics, and they would have caught every failure in the table two slides ago.**

# The shape of a lecture, and why

| Block | Why it is there |
|---|---|
| Where we are | so a lecture you missed is recoverable |
| The derivation | because the method does not make sense without it, and it outlasts the libraries |
| The method, with a worked example | real numbers from real data, reproducible from the repository |
| Failure conditions | stated as properties, never sprung as traps |
| The notebook | complete, correct, and printing every figure the slides state |

Every number on every slide of this course is produced by a script in the repository, and every figure a slide states is one its own notebook prints. Both are checked mechanically.

# The datasets, and why each

| | |
|---|---|
| California housing | regression, and a target with a cap in it |
| MNIST | a rare event carved out of a balanced corpus |
| Titanic | small, mixed types, and a real need for probabilities |
| CoverType | large enough that trees and ensembles differ |
| Olivetti | more dimensions than samples |
| Fashion-MNIST, CIFAR-10 | networks, then deep networks |
| Flowers102 | too few labels to train from scratch |
| COCO | several objects per image, and captions |
| Chicago transit | ordered rows, and strong seasonality |
| IMDb | free text |
| SciFact, MovieLens | ranking, and implicit feedback |

BEING EXPLICIT

# What this course did not cover

and where those gaps sit

# Why this slide exists

Because “I took a machine learning course” is read by other people as a claim about what you can do. You should be able to say precisely what it does and does not mean.

> **THE SCOPE, ONE LAST TIME**
>
> Chapters 1–16 of one book, thoroughly, on twelve applications. That is a large and useful slice of applied machine learning. It is not the field.

# Gaps: what this course did not cover

- **Reinforcement learning** — an agent acting, and learning from the consequences. A different problem shape entirely
- **Generative image models** — diffusion, and what it optimises
- **Causality** — every method here predicts; none of them tells you what happens if you *intervene*
- **Fairness and model governance** — who a model is worse for, and who is accountable when it is
- **Serving and drift** — the model in production, and how it decays quietly once the world moves

Named so you know they exist and roughly where they sit. None is examinable, and each is a course.

# The two you are most likely to need first

## Drift

Every number in this course was measured on data drawn like the training data. In production that stops being true, gradually and without warning, and the score you reported at launch is the last honest one anybody has.

Monitor the inputs, not just the outputs: the distribution moves before the accuracy does.

## Who it is worse for

Lecture 2 sliced the error by district and found the poorest tenth predicted worst. That slice was one line of code, and it is the difference between a system you can defend and one you cannot.

The habit generalises: always ask for whom the average is wrong.

# Gaps that are in the book, past our chapter

| Topic | Where it sits |
| --- | --- |
| Autoencoders, GANs, diffusion models | generation rather than prediction; the same encoders, a different objective |
| Reinforcement learning | learning from a reward signal instead of labelled examples — a different problem class entirely |
| Training and serving at scale | data pipelines, distributed training, deployment |

Nothing above is examinable. All three are the natural next chapters of the same book, and you now have the vocabulary for them.

# Gaps that are beyond the book entirely

| Topic | Why it matters, in one line |
| --- | --- |
| Causal inference | every model here answers “what is associated with what”; deciding an *intervention* needs different assumptions and different data |
| Uncertainty quantification | we produced one confidence interval, in Lecture 2. Calibrated predictive uncertainty is its own subject |
| Fairness and harm | we ran one fairness check. Who a system fails, and what that costs them, is not a metric you can pick off a list |
| Privacy | differential privacy, federated learning, and what a model memorises about its training data |
| Monitoring and drift | a model is fitted once and lives for years. Detecting that the world moved is a permanent job |
| Experimental design | an offline metric improving is not a business improving. That is an A/B test, and it is a statistics problem |

# And the part that moves fastest

- **Pretraining at scale.** We used four pretrained models today and trained none of them. What it takes to make one — the data, the cost, the failure modes — is not in this course
- **Instruction tuning and alignment.** The generator in the last section behaves the way it does because of a training stage we never discussed
- **Evaluating generated text.** We measured grounding because grounding is checkable. Measuring whether an answer is *good* is unsolved

> **THE TRANSFERABLE PART**
>
> Every one of those still fails in the ways this course taught you to look for: the wrong metric, a missing baseline, a test set that leaked, and a number quoted from one run.

# What you can measure without a ground truth

- **Consistency** — ask the same thing twice, in two wordings. Disagreement bounds reliability from above
- **Groundedness against the retrieved text** — mechanical, and it does not need a gold answer
- **Abstention on unanswerable queries** — construct them deliberately; this is the cheapest useful test of a RAG system
- **Human judgement on a sample** — expensive, and still the anchor everything else is validated against

> THE RULE, UNCHANGED SINCE LECTURE 1
>
> A metric with nothing to compare it to is decoration. That is as true of a generated answer as it was of an RMSE, and harder to honour.

# The five questions, for the last time

1. What touched the test set?
2. What was fitted, and on what?
3. What is the shape here?
4. What was dropped — rows, columns, NaNs? Count them
5. What is the default I did not ask for?

Question 3 caught a crash in Lecture 23. Question 5 caught two assistant failures in two consecutive lectures. They keep working.

# The four rules, unchanged since the first lecture

1. **You may not submit code you cannot explain line by line**
2. **Every number needs a baseline**
3. **Every model needs an ablation**
4. **Report the failure cases**

Rule 2 did the most work this application: the difference between “cosine 0.006” and “cosine 0.006, against 0.003 for unrelated pairs” is the whole of Lecture 23.

# What you can do now

1. Take a problem, frame it, and say what a correct answer would be *before* fitting anything
2. Choose a metric that matches the brief, and measure the trivial baseline first
3. Build the simplest thing that runs, inside a pipeline, evaluated on a protocol you fixed in advance
4. Read the failure: slice the error, look at the worst cases, and name the conditions under which the method breaks
5. Improve one thing at a time, and report the ablation rather than the best row
6. Write down what the number was measured on, every time

Six steps. Nothing in them is specific to a library, and none of them will be out of date in ten years.

# A last word on the assistant

You will build most of this with an assistant, and Lecture 1 set out how: specify, generate, *read*, test, verify — and steps three to five are yours.

> **WHAT HAS CHANGED BY LECTURE 24**
>
> You now know what to look for. “What was this measured on?”, “what is the baseline?”, “was anything fitted before the split?” — these are the questions an assistant will not ask itself, and they are precisely the ones that decide whether the number it hands you means anything.

**✗ The code was never the scarce part. Knowing whether to believe the output is, and that is what these twenty-four lectures were for.**

# The examination

## **Written** — two hours, closed book

- A · the derivations (40%)
- B · choosing a method for a stated situation (35%)
- C · reading results (25%)

## **Oral** — optional, seven to ten minutes

- Three questions drawn in front of you from the 120 exercises
- No notes, no computer
- You decide after seeing your written mark

The written is marked out of 30, pass at 18, capped at 27 on its own; the oral moves it by at most ±3 and cannot turn a pass into a fail. Everything in Chapters 1–16 and in the Part V notes is examinable; nothing marked *engineering* or *for context* is.

# How to keep learning

- **Reproduce something.** Take a paper's headline number and try to get it. You will learn more from the gap than from the paper
- **Keep the habit of the baseline.** It is the single cheapest defence against a wrong conclusion, and almost nobody does it
- **Read the evaluation section first.** If it does not say what was held out and how, the rest is decoration
- **Write the check before the model.** Every notebook in this course does, and it is why they still run

The libraries in these notebooks will be superseded. The derivations will not, and neither will the habit of asking what a number was measured on.

# Every notebook in the course

01 · What machine learning is, and how we will work

02 · The end-to-end project

03 · Classification and its metrics

04 · Training models

05 · Regularisation and the bias–variance trade-off

06 · Decision trees

07 · Ensembles and random forests

08 · Dimensionality reduction and unsupervised learning

09 · Neural networks, from the perceptron up

10 · PyTorch

11 · Training deep networks

12 · Convolutional networks

13 · Transfer learning

14 · Detection and segmentation

15 · Time series

16 · Recurrent networks

17 · Text

18 · Attention and transformers

19 · Information retrieval: the lexical foundation

20 · Information retrieval: dense retrieval

21 · Recommender systems: from ratings to factors

22 · Recommender systems: neural, and evaluated honestly

23 · Vision transformers and multimodal retrieval

24 · Generation, retrieval-augmented systems, and where this leaves you

The full index of all twenty-four is on the course site. Every one of them is ours, written from a specification, in the lecture.

# In the book

| Topic | Chapter |
|---|---|
| Softmax, cross-entropy, logits | 4, 15 |
| Transformer encoders and decoders; cross-attention | 15 |
| Vision transformers; text–image models; zero-shot | 16 |
| Image captioning | 16 |
| Random projection and concentration | 7 |

Retrieval-augmented generation, the modality gap, and the evaluation of generated answers are outside Chapters 1–16. They are here because the application needed them, and they are flagged as non-examinable.

# What we did today

1. Wrote captions for the entries that had none, and found what a caption buys and what it invents
2. Established that prompting is a hyperparameter, and chose one the way every other hyperparameter in this course was chosen
3. Joined a retriever to a generator, and located the failures in the *join* rather than in either half
4. Made the grounding check mechanical, because a citation that looks right and is not is exactly this course's subject
5. Filled 60 missing descriptions automatically and recovered R@1 from 0% to 61.7% on those entries
6. Grounded an assistant in the catalogue and moved its citation validity from 0% to 100%

# If you remember one thing

THE LINE

**A number is only as good as the question “measured on what?” can be answered.**

Every failure in this course — the leak, the training score, accuracy under imbalance, the random split on a series, the augmented validation set, the ungrounded generated answer — is a case where that question had an answer nobody had asked for.

Asking it costs seconds. Not asking it has cost, in these twenty-four lectures alone, an RMSE of zero, a 90% classifier that never fires, a network that does not train, and a forecast of the past.

NOT PRESENTED — FOR AFTERWARDS

# Exercises
# Lecture 24

five, in the style of the written paper

# Exercises — Lecture 24

1. A contrastive loss is written as `img @ txt.T / tau` on the output of `get_image_features`. State the defect, and the assertion that would have caught it. [5]

2. An auto-caption is generated for catalogue entries that have no description. State what it recovers, and the failure mode nothing in the evaluation detects. [5]

3. A retrieval-augmented system is measured by whether cited stock numbers exist. State what that measures, and what it does not. [4]

Solutions overleaf — there is no Lecture 25.

# Exercises — Lecture 24 (continued)

**4** In a retrieval-augmented pipeline the retriever's recall bounds the whole system. Explain why, and what that implies about where to invest. [4]

**5** Give the five rules the course ends on, and state the property the deck points out that all five share. [5]

Solutions overleaf — there is no Lecture 25.

THE FIVE SET AT THE END OF LECTURE 23

# Solutions
# Lecture 23

not part of this lecture

# Solutions — Lecture 23

**1.** Two encoders are trained separately, one on images and one on text. Explain why their cosine similarity…

**✓ Any rotation of the image space leaves every image–image cosine and the image loss unchanged, while changing every image–text cosine.**

- The objective cannot distinguish the rotated space from the original
- So it did not pick one, and the relative geometry is arbitrary

**2.** In $d$ dimensions two independent random unit vectors have a cosine with standard deviation $1/\sqrt{d}$.…

**✓ Near zero, with a spread of about 0.044. There is exactly one point at cosine $-1$, so most pairs cannot be there.**

- Unrelated means orthogonal in high dimensions, not opposite
- It is why a contrastive loss targets zero for a non-matching pair

# Solutions — Lecture 23 (2)

**3.** A contrastive loss is computed at $\tau = 1$ on cosine scores. State why it barely trains, and what $\tau\ldots$

**✓ Cosines span at most 2, so the softmax is nearly uniform. $\tau$ changes where the gradient goes, not which column is largest.**

- A range of 2 gives a ratio of at most $e^2$ across all competitors
- Temperature cannot change the accuracy of a fixed similarity matrix

**4.** Recall@10 is reported over 200 candidates. State the exact random baseline, and why it is computed rather…

✓ $10/200 = 5\%$.

- One relevant item ranked uniformly gives $P(\mathrm{rank} \le k) = k/n$
- A closed form needs no seed and carries no noise

# Solutions — Lecture 23 (3)

**5.** An entry with no written description scores zero on a text retrieval route. State whether this is a ranking…

## ✓ Structural — an entry with no text is not in a text index at all.

- No amount of re-ranking reaches a document that was never indexed
- The repair is another route, not a better scorer

ANSWERED HERE, BECAUSE THERE IS NO LECTURE 25

# Solutions
# Lecture 24

not part of this lecture

# Solutions — Lecture 24

**1.** A contrastive loss is written as img @ txt.T / tau on the output of get_image_features. State the defect, and…

**✓ The features are not normalised, so the product is not a cosine. Assert every row has unit norm.**

- The entries carry both magnitudes, so the temperature divides a quantity with no fixed scale
- Magnitude tracks length and token frequency, so long documents win by default

**2.** An auto-caption is generated for catalogue entries that have no description. State what it recovers, and the…

**✓ It recovers part of the lost recall. A caption that is wrong makes an entry findable under the wrong query.**

- Scoring generated captions against human captions of the same image is a friendly test
- Worse than unfindable, and no metric here sees it

# Solutions — Lecture 24 (2)

**3.** A retrieval-augmented system is measured by whether cited stock numbers exist. State what that measures, and…

## ✓ It measures grounding, not helpfulness.

- A cited entry can exist and still be a bad recommendation
- It is the cheap half, and saying which half was measured is the point

**4.** In a retrieval-augmented pipeline the retriever's recall bounds the whole system. Explain why, and what that…

## ✓ If the right entry is not retrieved, no generation recovers it.

- The generator can only work with what it is given
- The first stage sets the ceiling, exactly as in Lecture 20

# Solutions — Lecture 24 (3)

**5.** Give the five rules the course ends on, and state the property the deck points out that all five share.

**✓ Split before anything is fitted; compute the trivial baseline first; two numbers measured on different rows are not comparable; a difference smaller than the spread is not a difference; change one thing. None of them is mathematics.**

- Every one governs what a number may be compared with, not how a model is fitted — which is why none of them needed any
- They are the five that would have caught every row of the catalogue of silent failures, each of which runs to completion and raises nothing

APPLICAZIONI INFORMATICHE DEL MACHINE LEARNING

# Thank you

Eighteen derivations · twenty-four notebooks · thirteen datasets
Every number on every slide reproduced by a script in the repository

Questions, and then the examination